

**포트폴리오 수익률 분포의
비대칭적 의존성의 표본외 예측가능성
: Copula 분포함수에 의한 추정**

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The Out-of-Sample Predictability of Asymmetric Dependence of Portfolio Returns – The Multivariate Copula Distribution Function Approach

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〈Abstract〉

Armed with the copula distribution function that describes the asymmetric tail dependence, and the marginal distributions that capture the fat-tailed behavior, we estimate risk measures such as the Value-at-Risk and expected shortfall and evaluate whether those from the Gaussian copula function underestimate or overestimate true risk measures. We also investigate the impact of asymmetric tail dependence between the portfolio returns on the out-of-sample predictability of the returns. We fit the GPD as the two margins and a variety of copula functions in the extant literature in evaluating the risk measures. We compute the VaR and ES from the benchmark Gaussian copula model and the competing copula models and confirm that the benchmark model underestimate the levels of risk regardless of the measures of dependence. We use the out-of-sample predictability test to

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evaluate the performance of the competing copula models. We compare the out-of-sample predictability of the three copula-based competing models by calculating the out-of-sample log-likelihood. The Gaussian and the Student's t copula models are the representative symmetric copulas, while the Clayton copula is selected as the representative asymmetric copula model. According to the test statistics, the out-of-sample predictability of the Clayton and the Student's t copula models are superior to that of the Gaussian copula model. Overall, the Clayton copula model turns out to be the best out-of-sample forecasting copula model.

Keywords: Copula function, Asymmetric Dependence, Tail Dependence,
Out-of-sample forecast, Value-at-Risk, Expected Shortfall

JEL Classification: C13, C14, C15, C22

I . INTRODUCTION

The mean-variance analysis of portfolio is based on the symmetric behavior of the multivariate joint distribution of stock portfolio returns. The violation of this assumption is investigated by many authors. A voluminous literature has focused on measuring the temporal dependence between financial markets and assessing the impact on the risk measurement and the out-of-sample predictability of returns.

Patton (2006b) uses the copula function methodology to develop a time varying conditional joint distribution of exchange rate returns and prove the existence of asymmetric tail dependence structure in the returns. Bouyé et al. (2002) measure nonlinear dynamic dependence structure of non-Gaussian bivariate random variables and evaluate the impact of dependence dynamics on the financial return predictability. Patton (2004) investigates the impact of non-elliptical return distribution and the existence of asymmetric dependence of asset returns on asset allocation decisions and concludes that the model with skewness and asymmetric dependence performs better than the benchmark model with bivariate normal distribution assumption in the out-of-sample predictability. Chen et al. (2004) hypothesize the dependence structure as the Gaussian copula and the Student's t copula and find significant evidence against the Gaussian copula and for the Student's t copula using exchange rate returns data. Patton (2006b) specifies the asymmetric dependence structure to construct the joint distribution functions, while he explores the possibility of misspecifying the marginal distributions. Bouyé and Salmon (2009) employ the copula-based quantile regression method to estimate tail dependence measure and prove that the bivariate joint distribution function for exchange rates is dynamically independent.

In this paper we investigate the impact of asymmetric tail dependence between the portfolio returns on the risk measures and on the out-of-sample predictability. The impact of asymmetric dependence on the risk measures and on the out-of-sample predictability is compared using a variety of bivariate copula models. We compute the risk measures such as the Value-at-Risk and expected shortfall from the copula distribution function that describes the asymmetric tail dependence of data best and compare the risk measures with those from the Gaussian copula. Also, we compare the out-of-sample predictability of various bivariate copula models.

The aims of this paper are three folds. Firstly, we use various tests whether there is asymmetric dependence in the data. Secondly, we use the copula distribution function and the marginal distribution function to separate out the dependence structure of the joint distribution of the portfolio returns. Thirdly, we estimate the copula distribution function that describes the asymmetric tail dependence and predicts the data out-of-sample. Fourthly, we compare the risk measures such as the Value-at-Risk and expected shortfall and determine whether those from the Gaussian copula function underestimate or overestimate true risk measures.

The remainder of this paper is arranged as follows. We proceed with characterizing the marginal distributions and the dependence structure of the two stock index returns. The generalized Pareto distribution and the skewed t distribution are employed for the marginal distributions, and a variety of copula models are used for the specification of the dependence structure. Armed with the copula distribution function that describes the asymmetric tail dependence, we estimate the risk measures such as the Value-at-Risk and expected shortfall and evaluate whether

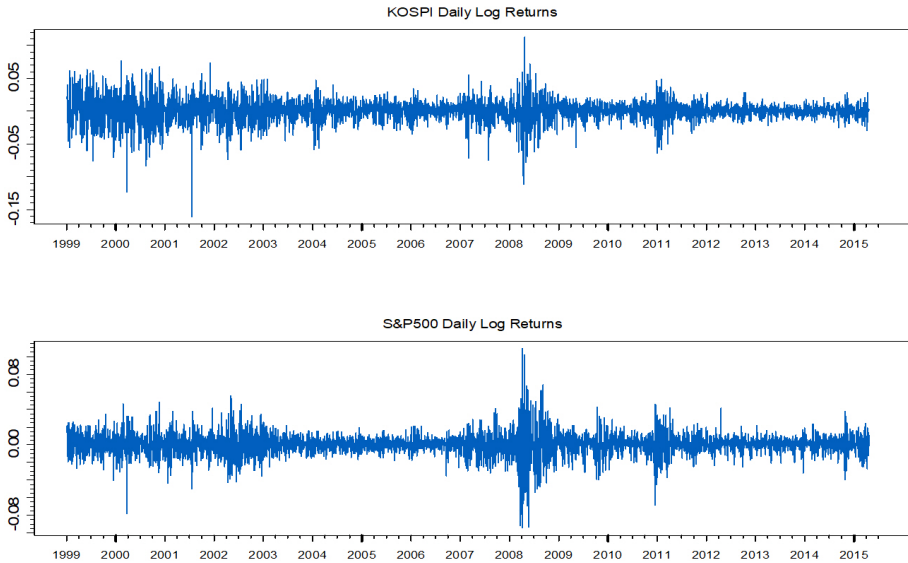
those from the Gaussian copula function underestimate or overestimate true risk measures. We also investigate the impact of asymmetric tail dependence between the portfolio returns on the out-of-sample predictability. Section 3 concludes the discussion.

II. MODELLING AND ESTIMATION

1. Marginal Distributions

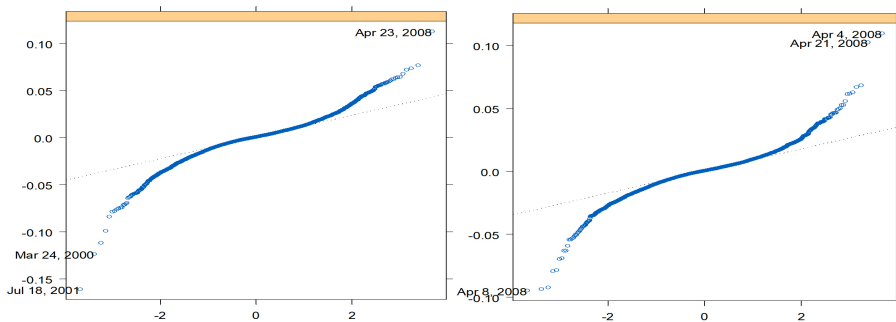
We use the daily log returns on the KOSPI 200 and S&P 500 ranged from January 4, 1999 to February 26, 2016, yielding 4,109 observations. Figure 1 depicts the log returns on the two stock indices. The extreme volatilities are exhibited during the years from 1999 to 2001 and around 2008 for the KOSPI and during the years around 2008 for the S&P 500. To figure out whether the actual returns deviate from the standard normal distribution, the qq-plots against the standard normal distribution are presented in Figure 2. As we observe from the figure, the marginal distribution is required to include the characteristics of fat-tailed distribution.

〈Figure 1〉 Daily Log Returns on the KOSPI 200 and S&P 500



Note: This figure presents the daily log returns on the KOSPI 200 and S&P 500 during the period from January 4, 1999 to February 26, 2016, resulting in 4,108 observations. The extreme volatilities are exhibited during the years from 1999 to 2001 and around 2008 for the KOSPI and during the years around 2008 for the S&P 500.

〈Figure 2〉 QQ-plots of the Returns on the KOSPI 200 and S&P 500 against the Normal Distribution



Note: These figures present the qq-plots of the daily log returns on the KOSPI 200 and S&P 500 against the normal distribution as the reference distribution. The curve down to the left and the curve up to the right explain the fat-tailed behavior in each extreme.

Under the bivariate normality, the correlation coefficient between the two index returns is calculated at 0.18. However, the null hypothesis of normality is rejected with the p-value of 0.00 (Jarque-Bera test statistic = 6,608.59) for the KOSPI 200 and 0.00 (Jarque-Bera test statistic = 9,584.09) for the S&P 500. The bivariate normal distribution is hardly a good characterization of the two returns series. Figure 3 shows two scatter plots of the actual returns and the simulated returns calibrated to the actual data. Not much of a difference between the two scatter plots can be found in the middle of the joint distribution, except that the actual returns are more densely populated than the simulated ones. Specifically, the simulated bivariate distribution does not seem to describe the joint behavior of the actual returns in the extreme two tails. We can infer from the dependence structure in the tails that the dependence measure may be asymmetric and time-varying. To confirm this assertion, we estimate the generalized Pareto distribution (GPD), $G_{\xi, \beta(u)}(y)$.

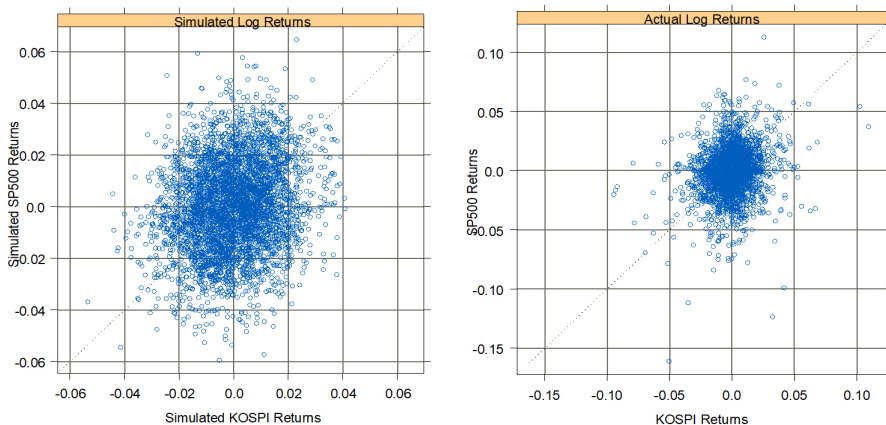
$$G_{\xi, \beta(u)}(y) = \begin{cases} 1 - \left(1 + \left(\frac{\xi y}{\beta(u)}\right)^{-\frac{1}{\xi}}\right), & \text{for } \xi \neq 0, \beta(u) > 0 \\ 1 - \exp\left(-\frac{y}{\beta(u)}\right), & \text{for } \xi = 0, \beta(u) > 0 \end{cases},$$

defined for $y \geq 0$ when $\xi \geq 0$ and $0 \leq y \leq \frac{-\beta(u)}{\xi}$ when $\xi < 0$.

The GPD is useful in modeling the tails of the marginal distribution. The empirical distribution function is coupled with the GPD in modeling the center of the distribution, which results in the semiparametric GPD. The semiparametric GPD models for the two returns are estimated using the maximum likelihood estimator. For the KOSPI 200, 21.52% of the observations are selected for the upper tail parameter estimation, and

19.52% of the observations are used for the lower tail parameter estimation. The GPD parameters of the upper tail are estimated at $\xi = 0.08$ with standard error of 0.039 (t-value=2.04), and $\beta = 0.01$ with standard error of 0.001 (t-value=19.54). The GPD parameters of the lower tail are estimated at $\xi = 0.07$ with standard error of 0.035 (t-value=2.07), and $\beta = 0.01$ with standard error of 0.001 (t-value=20.02). For the S&P 500, 15.30% of the observations are selected for the upper tail parameter estimation, and 15.66% of the observations are used for the lower tail parameter estimation. The GPD parameters of the upper tail are estimated at $\xi = 0.16$ with standard error of 0.046 (t-value=3.44), and $\beta = 0.01$ with standard error of 0.001 (t-value=16.33). The GPD parameters of the lower tail are estimated at $\xi = 0.11$ with standard error of 0.041 (t-value=2.76), and $\beta = 0.01$ with standard error of 0.001 (t-value=17.61).

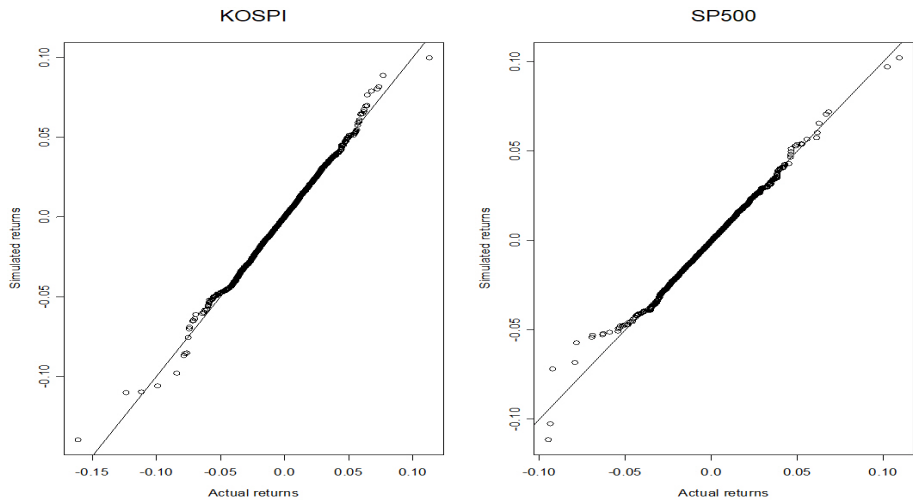
〈Figure 3〉 Actual and Simulated Daily Log Returns on the KOSPI 200 and S&P 500



Note: The lower panel of figure 3 presents the scatter plot of the actual daily log returns on the KOSPI 200 and S&P 500 and the upper panel shows that of simulated returns calibrated to the actual returns with the bivariate normal distribution assumption.

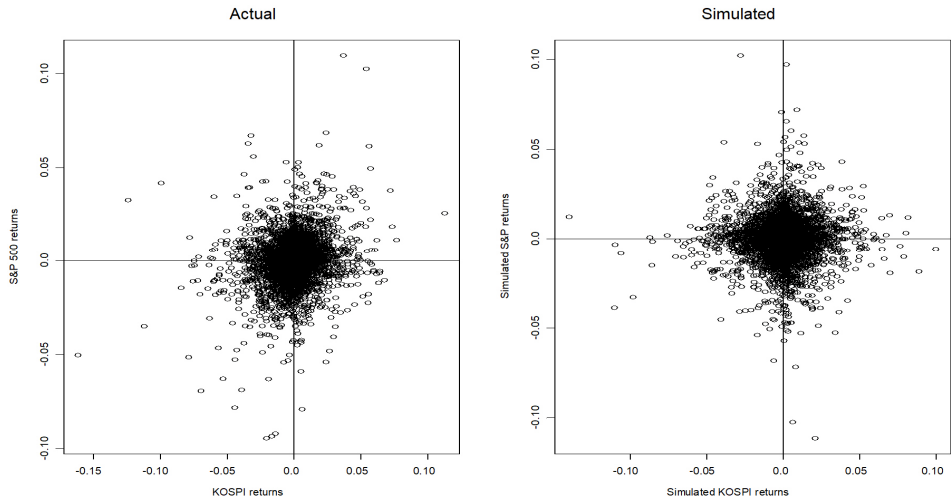
We compare the actual returns to the simulated returns calibrated to the GPD tail estimation results as shown in Figure 4. The simulated returns describe the actual returns in isolation, except in a few observations that the qq-plot curves down and up to the left and up to the right for the KOSPI and that the qq-plot curves up to the left for the S&P 500. The simulated bivariate returns are presented in Figure 5. We compare the simulated bivariate returns calibrated to the GPD estimation results in Figure 5 to the simulated bivariate normal returns in Figure 3. The simulated returns calibrated to the GPD estimation results replicate the actual scatter plot in Figure 3 more closely, however, the correlation coefficient between the simulated KOSPI 200 and S&P 500 returns is calculated at 0.0057. We can conclude that the simulated bivariate returns from the GPD model reproduces the tail related behavior at the upper and lower extreme well, but not the dependence structure of the actual returns. To address the dependence structure of the joint distribution, we introduce the copula methodology.

〈Figure 4〉 QQ-plot of Each Return Against the GPD



Note: The left panel presents qq-plot of actual daily log return on the KOSPI 200 against the simulated returns calibrated to the GPD as the reference distribution. The right panel shows qq-plot for the S&P 500.

〈Figure 5〉 Scatter Plot of the Simulated Returns Fitted to the GPD



Note: The simulated returns calibrated to the GPD estimation results are shown. Compared to the simulated returns calibrated to the bivariate normal distribution in Figure 3, this replicates the actual data closely, however, the linear correlation between the two component returns is computed at 0.0057 which is much smaller than that between the actual returns.

We assume that the conditional marginal distribution is set up for the conditional mean and conditional variance to be dependent upon time:

$$Y_{it} = \mu_i(Z_{t-1}) + \sigma_i(Z_{t-1})\varepsilon_{it}, \text{ for } i = 1, 2, \text{ where } Z_{t-1} \in F_{t-1}.$$

$$\varepsilon_{it}|F_{t-1} \sim F_i(0, 1) \text{ for all } t.$$

Based on the BIC, we choose the AR(2) model for the KOSPI 200 and the AR(0) model for the S&P 500. The estimation results for the conditional mean are presented in the second panel of Table 1. In the table, the estimated parameters of the GJR-GARCH model of order (1,1) for the conditional variance are shown. As an alternative to the generalized Pareto distribution, the Normal, skewed t, and the empirical distribution function of the form, $\hat{F}_i(\varepsilon) \equiv \frac{1}{T+1} \sum_{t=1}^T 1\{\hat{\varepsilon}_{it} \leq \varepsilon\}$ are employed as the conditional marginal distribution function.

We fit the skewed t distribution to the standardized residuals and present the results in Table 1. We report the Kolmogorov-Smirnov (KS) and Cramer-von Mises (CvM) goodness-of-fit tests results. According to the p-values shown in the last two rows, we fail to reject the null hypothesis that the two index returns are distributed to the skewed t. We confirm that the skewed t distribution fits the estimated standardized residuals best due to Kolmogorov-Smirnov(KS) and Cramer-von Mises(CvM) goodness-of-fit test, and can appropriately be used for the conditional marginal distribution. Figure 6 presents the fitted density laid over the histograms of the data. The qq-plots against the empirical distribution shows a few deviations from the empirical distribution in the left tail. As is evidence by the qq-plot, the skewed t distribution can justifiably

be used as the conditional marginal distribution.

〈Table 1〉 Descriptive Statistics

	KOSPI 200	SP 500
Mean	0.000	0.000
Standard Deviation	0.017	0.013
Skewness	-0.552	-0.187
Kurtosis	9.098	10.476
Correlation(linear/rank correlation)	0.180	0.157

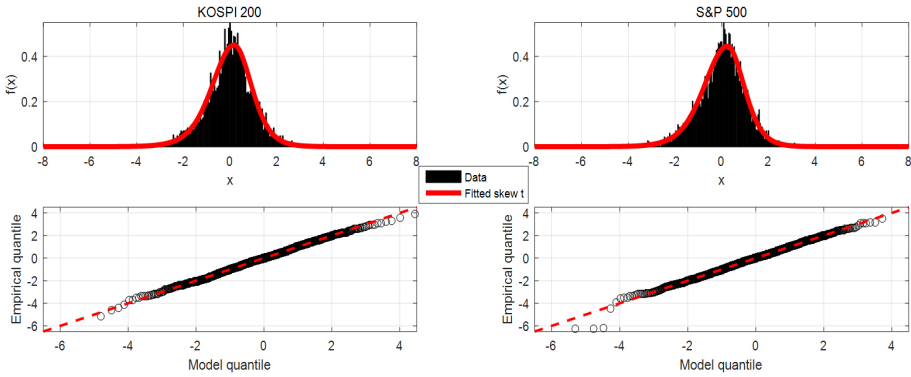
Conditional mean	KOSPI 200	SP 500
ϕ_0	0.0003	0.0001
ϕ_1	0.0263	-
ϕ_2	-0.0272	-

Conditional variance	KOSPI 200	SP 500
ω	0.0000	0.0000
α	0.0348	0.0000
δ	0.0711	0.1580
β	0.9272	0.9030

Skewed t density	KOSPI 200	SP 500
ν	7.784	8.731
λ	-0.112	-0.156
Kolmogorov Smirnov p-value	0.100	0.204
Cramer-von Mises p-value	0.109	0.243

Note: The descriptive statistics are shown in the first panel. The parameter estimates of the conditional mean and the conditional variance models are presented in the second and the third panel. We also report the skewed t density fit in the last panel. The p-values of the Kolmogorov-Smirnov(KS) and Cramer-von Mises(CvM) tests are reported in the last panel.

〈Figure 6〉 The QQ-Plots against the Fitted Distribution



Note: The figure shows the fitted skewed t density (the bold lines at the upper panel) compared to the actual data (histograms). The qq-plots against the empirical distribution shows a very few deviations from the empirical density.

2. Dependence Structure

In the previous analysis, we show that the skewed t distribution fits the data remarkably well. In this section, we investigate a variety of dependence measures of the random variables from the joint distribution functions that are scale invariant to strictly increasing transformations of the marginal distributions. The rationale behind this is that, if the measures of dependence and the copula distribution functions are both invariant to increasing transformations of the marginal distribution functions, the dependence measures can be represented by the function of the copula between the variables. We try to achieve the scale invariant dependence measures that can be characterized by the copula and not by the marginal distributions of the random variables.

To that end, we estimate the quantile dependence in equation (1) which can be used to represent the quantile-varying asymmetric dependence.

$$\lambda^q = \begin{cases} \Pr[U_{1t} \leq q | U_{2t} \leq q], & 0 < q \leq 1/2 \\ \Pr[U_{1t} > q | U_{2t} > q], & 1/2 < q < 1 \end{cases} \quad (1)$$

$$= \begin{cases} \frac{C(q, q)}{q}, & 0 < q \leq \frac{1}{2} \\ \frac{1 - 2q + C(q, q)}{1 - q}, & \frac{1}{2} < q < 1 \end{cases}.$$

$$\hat{\lambda}^q = \begin{cases} \frac{1}{Tq} \sum_{t=1}^T 1\{U_{1t} \leq q, U_{2t} \leq q\}, & 0 < q \leq 1/2 \\ \frac{1}{T(1-q)} \sum_{t=1}^T 1\{U_{1t} > q, U_{2t} > q\}, & 1/2 < q < 1 \end{cases}. \quad (2)$$

We also estimate the tail dependence in equation (3). The tail dependence can be derived from equation (1) at the limit, that is as $q \rightarrow 0$ or $q \rightarrow 1$, and it measures how closely the extreme events in both markets are interrelated.

$$\lambda^L = \lim_{q \rightarrow 0^+} \frac{C(q, q)}{q}. \quad (3)$$

$$\lambda^U = \lim_{q \rightarrow 1^-} \frac{1 - 2q + C(q, q)}{q}.$$

Alternatively, we can estimate the lower and upper tail dependence non-parametrically due to Frahm et al. (2005):

$$\hat{\lambda}^L = 2 - \frac{\log(1 - 2(1 - q^*) + T^{-1} \sum_{t=1}^T 1\{U_{1t} \leq 1 - q^*, U_{2t} \leq 1 - q^*\})}{\log(1 - q^*)} q^* \approx 0. \quad (4)$$

$$\hat{\lambda}^U = 2 - \frac{\log(T^{-1} \sum_{t=1}^T 1\{U_{1t} \leq 1 - q^*, U_{2t} \leq 1 - q^*\})}{\log(1 - q^*)} q^* \approx 0.$$

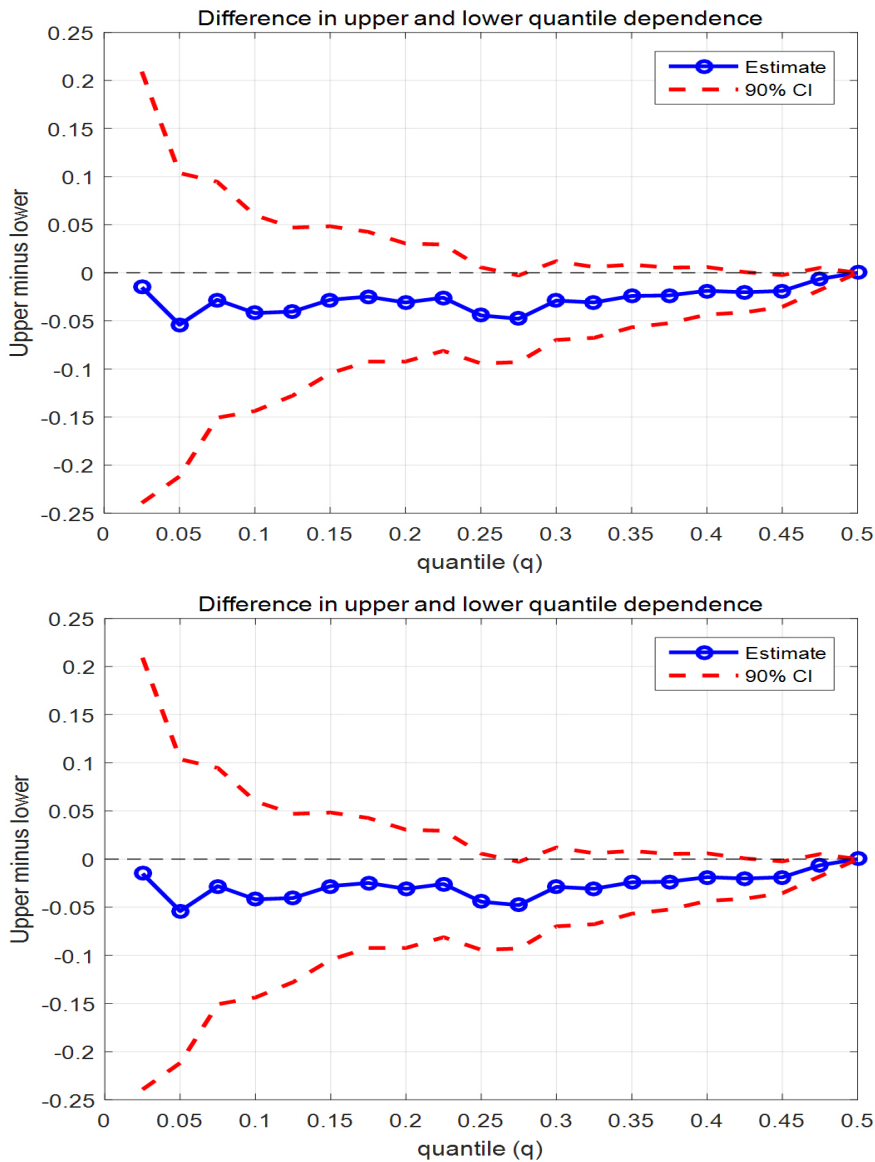
As the benchmark dependence measure, we get the sample rank

correlation, using $\hat{\rho} = \frac{12}{T} \sum_{t=1}^T U_{1t} U_{2t} - 3$, as 0.1856 with the 90% confidence interval of [0.1598, 0.2114]. The rank correlation represents the sign of dependence, thus we can interpret the positive sample rank correlation as representing positive interdependence between the two stock markets.

Figure 7 presents the quantile dependence from equation (2) for a variety of quantile. As we can see from the upper panel of Figure 4, the dependence measure between negative returns in the lower quantile are higher than that between positive returns in the upper quantile. The lower panel of Figure 7 presents the difference between the upper and lower quantile as $\hat{\lambda}^U - \hat{\lambda}^L$. The asymmetric dependence is evidenced in both figures since, for all the quantiles, the lower quantile dependence is higher than the upper quantile.

To clarify this observation, we use the statistic to test the null hypothesis of symmetric quantile dependence. The symmetric quantile dependence can be stated as $\lambda^q = \lambda^{1-q} \forall q \in [0, 1]$, where λ^q is from the quantile dependence equation (1). We implement the test for asymmetric quantile dependence with $q \in \{0.025, 0.05, 0.10, 0.975, 0.95, 0.90\}$. The null hypothesis, $R\lambda = 0$ where $R \equiv [I_p : -I_p]$ is tested against the alternative hypothesis, $R\lambda \neq 0$. The test statistic is calculated as -0.0823 with the p-value of 0.5530. Thus, we fail to find the evidence on asymmetric quantile dependence. The test results show that the lower tail has stronger tail dependence than the upper tail dependence, however, the difference between the two is not statistically significant.

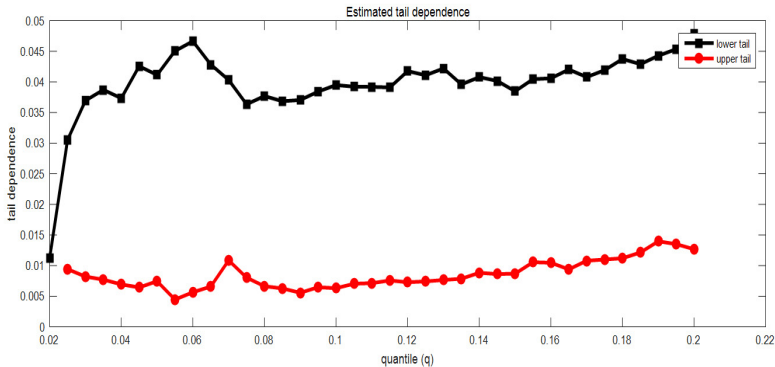
〈Figure 7〉 Quantile Dependence for KOSPI and S&P 500 Standardized Residual



Note: Quantile dependence between the standardized residuals from the two index returns are presented in the upper panel. The differences in the quantile dependence (upper minus lower) are shown in the lower panel. As evidenced from the confidence interval, however, the differences in upper and lower quantile dependence are not different from zeros.

Next, we measure the tail dependence from equation (4). When the tail dependence is estimated, it is important to balance the trade-off relationship between efficiency and unbiasedness in the process of selecting the quantile threshold. By selecting a quantile close enough to 0.5 from both tails, we can obtain a statistically efficient tail dependence measure. Alternatively, we can have a unbiased tail dependence measure by choosing a quantile close enough to each extreme and estimating the dependence measure with observations pertained to the ‘true’ tail. We use equation (4) due to Frahm et al. (2005). The estimate of lower tail dependence is 0.0429 with the 90% confidence interval of [0,0.3244]. The estimate of upper tail dependence is 0.0304 with the 90% confidence interval of [0,0.3257]. Figure 8 shows the lower and upper tail dependence measures with reference to a variety of cut-off threshold quantile values, q .

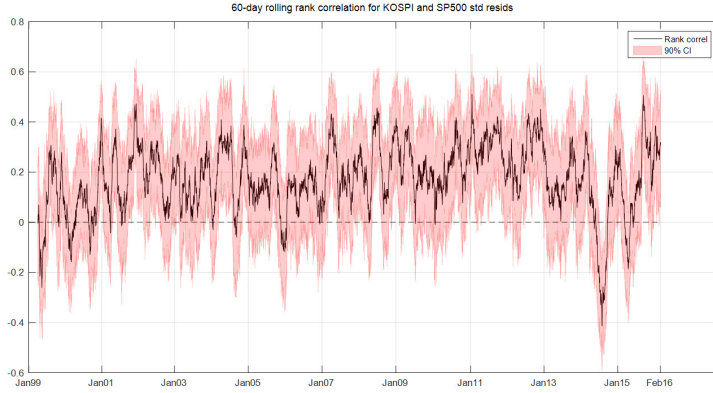
〈Figure 8〉 Lower and Upper Tail Dependence Coefficient Estimate



Note: The dependence measures in relation to a variety of quantile values are presented in the figure. The lower tail dependence is higher than the upper tail dependence for the cut-off threshold values, however, the difference between the two is statistically insignificant.

Lastly, we test for the existence of a time-varying dependence. To that end, the 60-day rolling window estimation procedure of rank correlation coefficients is presented in Figure 9. We choose the level shift in the rank correlation coefficients at $i^*/T \in \{0.15, 0.50, 0.85\}$ point in time, which corresponds to the dates of August 3, 2001, August 9, 2007 and August 1, 2013. The truncation of a whole sample is due to Andrews (1993) and this secures enough of observations to calculate the rank correlation with the pre-break and the post-break sample, respectively. The p-value of 0.026 from the date of August 3, 2001 instructs us to reject the null hypothesis of a one-time break in the rolling rank correlation at the 5% significance level, so we cannot choose the date of August 3, 2001 as the timing of the break. For other dates assumed for the timing of the break in rank correlation, we fail to reject the null hypothesis. As is evidenced in Figure 9, we find that the rank correlation experiences the level shifts on August 9, 2007 and August 1, 2013. The estimation results of an autoregressive model for $U_{(1,t)}U_{(2,t)}$ of the form $U_{1,t}U_{2,t} = \alpha_0 + \sum_{i=1}^p \alpha_i U_{1,t}U_{2,t-i} + \epsilon_t$ presented in Table 2 show that the autocorrelation coefficients for lags 1 and 5 are significantly different from zero at the 1% significance level. From the analysis of this section, we can conclude that the two stock index returns have the asymmetric quantile and tail dependence that have to be reflected in the time-varying copula models. This is what we are trying to turn to in the following section.

〈Figure 9〉 60-Day Rolling Window Estimation Results of Rank Correlation



Note: The figure shows a plot of rank correlation with bootstrap standard errors using a 60-day rolling window. The rank correlation seems to experience the level shifts on August 9, 2007 and August 1, 2013.

〈Table 2〉 Tests for Time-Varying Dependence

	0.15	0.5	0.85	AR(1)	AR(5)	AR(10)
p-value	0.026	0.988	0.208	0.000	0.007	0.056

Note: The table presents the test results for time-varying dependence. The p-value of 0.026 from the date of August 3, 2001 shows that we reject the null hypothesis that there exists a one-time break in the rank correlation at the 5% significance level. On the other hand, we fail to reject the null hypothesis for other dates assumed. The rank correlation seems to experience the level shifts on August 9, 2007 and August 1, 2013.

3. Copula Distribution Function

We define that X and Y are random variables with their joint and marginal distribution functions, $F_{xy}(x,y)$, $F_x(x)$ and $F_y(y)$, respectively. The random variables are independent if $F_{xy}(x,y) = F_x(x)F_y(y)$. A bivariate copula distribution function $C(F_x(x), F_y(y))$ is a distribution function on $I^2 = [0,1] \times [0,1]$ with the uniform distribution function as

marginal distribution functions. That is, $F_{xy}(x,y) = C(F_x(x), F_y(y)) = C(u,v)$, $u = F_x(x)$ and $v = F_y(y)$, for all $x,y \in [-\infty, \infty]$ according to the Sklar's theorem. Also, let $F_{xy}(x,y)$ be a joint distribution function with continuous marginal distributions F_x and F_y , then there exists a unique copula $C(F_x(x), F_y(y))$. So, the joint distribution function can be separated out into the marginal distributions and the copula. That is, $F_{xy}(x,y) = C(F_x(x), F_y(y)) = C(u,v)$, $u = F_x(x)$ and $v = F_y(y)$. With this specification of the copula distribution function, the maximum likelihood function can be established as $\ln f_{xy}(x,y) = \ln f_x(x) + \ln f_y(y) + \ln c(u,v)$, where u and v are the probability integral transforms of x and y and are uniformly distributed marginal distributions on $[0,1]$.

We can also infer from the Sklar's theorem that the copula distribution function is invariant to strictly increasing transformations of the marginal distributions of F_x and F_y . As a result, the dependence structure of the two random variables are invariant to strictly increasing transformations of the marginal distributions. Since the measures of dependence and the copula distribution functions are both invariant to increasing transformations of the marginal distribution functions, the dependence measures can be represented by the copula distribution function between the variables. In this sense, the parameter estimates of the copula function are interpreted as the measures of dependence as in equations (8) and (9) below.

(1) Parametric Copula Distribution Function

We define the parametric conditional mean and variance specification and the standardized residuals as follows:

$$\begin{aligned}
 E[Y_{it}|F_{t-1}] &\equiv \mu_i(Z_{t-1}, \alpha^*), & Z_{t-1} &\in F_{t-1} \\
 V[Y_{it}|F_{t-1}] &\equiv \sigma_i^2(Z_{t-1}, \alpha^*) \\
 \epsilon_{it} &\equiv \frac{Y - \mu_i(Z_{t-1}, \alpha^*)}{\sigma_i(Z_{t-1}, \alpha^*)}
 \end{aligned}$$

For the conditional distribution of ϵ_{it} , we take a parametric and a nonparametric case into account which result in different inference procedures for the copula parameters. Then, the parametric copula model is specified as follows:

$$\begin{aligned}
 U_{it} &\equiv F_i(\epsilon_{it}), & i &= 1, 2, \dots, n \\
 U_t &\equiv [U_{1t}, \dots, U_{nt}]' | F_{t-1} \sim \begin{cases} iid C(\gamma) \\ C(\delta_t(\gamma^*)) \end{cases}
 \end{aligned}$$

where δ_t is the copula parameter, and $C(\delta_t(\gamma^*))$ represents the time-varying copula. We can estimate $\theta^* \equiv [\alpha^{*'}, \gamma^{*'}]'$ using the maximum likelihood estimation procedure, where α^* is the parameters of the marginal distributions, and γ^* is the copula parameters.

(2) Empirical Results

We use the MLE of the log-likelihood to estimate time-varying copula models:

$$\log L(\theta|q) = \frac{1}{T} \sum_{t=1}^T l_t(\theta|q), \tag{5}$$

$$\text{where } l_t(\theta|q) = \delta_{1t}\delta_{2t}\log c(\tilde{U}_t, \tilde{V}_t; \theta) + \delta_{1t}(1 - \delta_{2t})\log \frac{\partial C(\tilde{U}_t, \tilde{V}_t; \theta)}{\partial u} \\ + (1 - \delta_{1t})\delta_{2t}\log \frac{\partial C(\tilde{U}_t, \tilde{U}_t; \theta)}{\partial v} + (1 - \delta_{1t})(1 - \delta_{2t})\log C(\tilde{U}_t, \tilde{V}_t; \theta).$$

We denote that $\tilde{U}_t = \max[U_t, q]$, $\tilde{V}_t = \max[V_t, q]$, $\delta_{1t} = 1\{U_t > q\}$, $\delta_{2t} = 1\{V_t > q\}$. Also, δ_{1t} and δ_{2t} represent the uncensored values. With upper and lower tail copula parameters, $\hat{\theta}^L$ and $\hat{\theta}^U$, the upper and lower tail dependence coefficients can be computed as:

$$\hat{\lambda}^L = \lim_{q \rightarrow 0^+} \frac{C^L(q, q; \hat{\theta}^L)}{q} \quad (6)$$

$$\hat{\lambda}^U = \lim_{q \rightarrow 1^-} \frac{1 - 2q + c^U(q, q; \hat{\theta}^U)}{1 - q} \quad (7)$$

We have three classes of copula functions, the Gaussian (benchmark) copula, the extreme value copula, and the Archimedean family copula. The extreme value copula class includes the Gumbel copula, the Galambos copula, the Hüsler and Reiss copula, the Twan copula, and the BB5 copula. The Archimedean copula class consists of the Clayton copula, the Frank copula, the Joe copula, and the Joe-Clayton copula. In this paper, we employ a variety of parametric copula models out of the candidates. The Clayton copula can be represented by the functional form of $C(u, v; \delta)$, where $\delta > 0$ and has the lower tail dependence parameter $\lambda^L = 2^{-\frac{1}{\delta}}$ and the upper tail dependence parameter $\lambda^U = 0$. The parameter δ denotes the dependence measure between the two marginal distribution functions. The Clayton copula model is a good

candidate to reflect the joint negative returns in the extreme quantile.

The Frank copula has parameters describing both positive (when $\delta \in (0, 1)$) and negative (when $\delta > 1$) dependence structures and has the functional form of $C(u, v; \delta)$. The Clayton and the Frank copula have a single parameter δ which represents the dependence measure. The BB3 copula is a function of lower (δ) and upper (θ) tail dependence and has the form of $C_{\delta, \theta}(u, v; \theta)$. The Joe-Clayton copula is a function with lower (δ) and upper (θ) tail dependence and has the form of $C_{\delta, \theta}(u, v)$. The Joe-Clayton copula is well suited to identify dependence structure in the extreme quantile on both sides. The parametric estimates of the lower (λ_L) and upper (λ_U) tail dependence measures are obtained by $\lambda_L = 2^{-\frac{1}{\delta}}$ and $\lambda_U = 2 - 2^{\frac{1}{\theta}}$, respectively. From the lower (λ_L) and upper (λ_U) tail dependence measures, asymptotic behavior in the extreme quantiles on both sides can be discovered. The lower tail dependence measure is estimated at 0.0143 and the upper tail dependence at 0.1150 from the parameter estimates.

Genest and MacKay (1986) and Bouyé et al. (2002) show that Kendall's τ and Spearman's ρ can be computed from the Archimedean generator. That is, the Kendall's τ is

$$\tau = 4 \int_0^1 \int_0^1 C(u, v) dC(u, v) - 1 = 4E(C(U, V)) - 1 \quad (8)$$

and the Spearman's ρ is

$$\begin{aligned} \rho &= 12 \int_0^1 \int_0^1 C(u,v) du dv - 3 = 12 \int_0^1 \int_0^1 uv dC(u,v) - 3 \\ &= 12 E(UV) - 3 = \frac{E(UV) - 1/4}{1/12} = \frac{E(UV) - E(U)E(V)}{\sqrt{Var(U)}\sqrt{Var(V)}} \end{aligned} \tag{9}$$

Table 3 shows the estimation results of the parametric copula models. Based on the information criteria, the Gumbel copula and the Clayton copula from each class are evaluated to minimize the information criteria. Table 4 reports the copula model estimation results which can be used to calculate the measure of lower and upper tail dependence.

〈Table 3〉 Log-Likelihood and Information Criteria Values from the Copula Models

Copula	Log-likelihood	AIC	BIC	HQ
Gaussian	59.79	-117.58	-111.26	-115.34
Gumbel	75.92	-149.84	-143.52	-147.60
Galambos	64.39	-126.79	-120.48	-124.56
Husler Reiss	59.20	-116.41	-110.09	-114.17
Frank	54.87	-107.75	-101.43	-105.51
Clayton	67.47	-132.94	-126.62	-130.70
Joe	58.31	-114.63	-108.31	-112.39

Note: This table shows the estimation results of the parametric copula models. Based on the information criteria, the Gumbel copula and the Clayton copula minimize the information criteria..

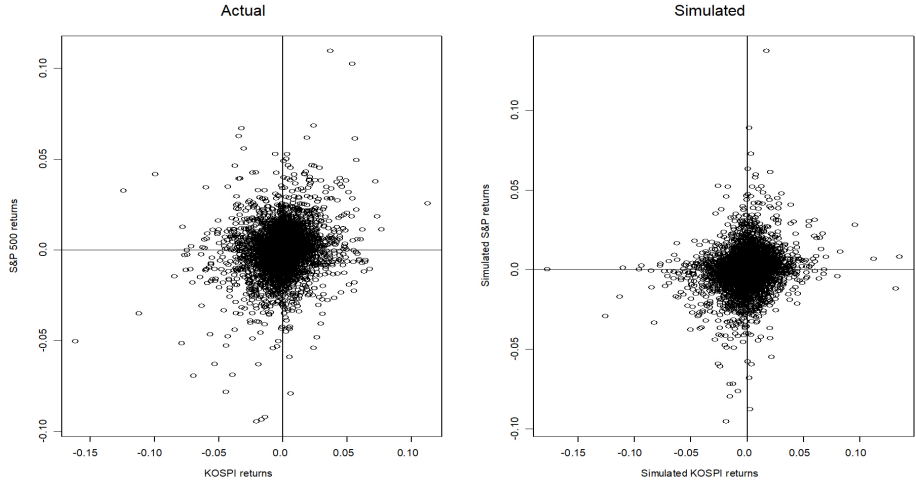
〈Table 4〉 Dependence Measure Coefficient from the Copula Models

Copula	parameter	estimate	std. Error	t-value
Gaussian	δ	0.1693	0.0149	11.3352
Gumbel	δ	1.1178	0.0119	94.3265
	θ	1.0934	0.0161	67.8964
BB7 (Joe-Clayton)	δ	0.1633	0.0215	7.6091
Frank	δ	1.0207	0.0974	10.4761
Clayton	δ	0.2105	0.0209	10.0765
Joe	θ	1.1381	0.0166	68.4406

We can estimate the empirical copulas and compute the non-linear dependence measures. For example, the Spearman's ρ and the Kendall's τ are obtained from the empirical copulas as in equation (3). The estimated GPD functions are used to form the integral transforms of the marginal distributions, and the empirical copula can be computed. The Kendall's τ and the Spearman's ρ are estimated at 0.108 and 0.157, respectively.

Figure 10 shows a scatter plot of simulated bivariate returns from the calibrated semiparametric GPD margins to the data. The GPD margin represents the tail behavior in each return, however, the dependence structure between the two returns is neglected in the simulation. We combine the two GPD margins and the normal copula with the correlation coefficient calibrated to the actual data in the simulation process.

〈Figure 10〉 Scatter Plots of the Simulated Returns Calibrated to the GPD Margins and Gaussian Copula



Note: The scatter plot of the simulated returns calibrated to the GPD margins and a normal copula with a correlation coefficient of 0.18 are shown in the right panel. Compared to the simulated returns calibrated to the bivariate normal distribution in Figure 3, this replicates the actual data closely with the linear correlation between the two component returns is computed at 0.174 which is similar between the actual returns. Compared to the scatter plot of the actual returns, however, the simulated returns are more densely populated in the middle of the distribution and do not replicate the actual returns at the extreme quantile.

〈Table 5〉 Estimation Results for the Tail Dependence

	Nonparametric	Parametric	
	Eq. (4), Frahm et al. (2005)	Gumbel copula	Student t copula
Lower tail dependence when $q \rightarrow 0^+$			
Estimate	0.0429	0.0193	0.0138
90% CI	[0,0.3101]	[0.0002,0.0526]	[0.0001,0.0250]
Upper tail dependence when $q \rightarrow 1^-$			
Estimate	0.0304	0.0099	0.0075
90% CI	[0,0.3135]	[0.0001,0.0097]	[0.0001,0.0218]
p-value for $\lambda^L = \lambda^U$	0.5390	0.2600	0.4600

Note: The numbers are the tail dependence estimates. The numbers in parentheses are the confidence intervals at the 10% significance level. From the p-values for the null hypothesis of $\lambda^L = \lambda^U$, we fail to reject that the lower tail dependence is

statistically equal to the lower tail dependence.

Table 5 reports the estimates of the tail dependence measures and the parametric copula functions. An iidbootstrap procedure is used to construct the 90% confidence intervals on these estimates. From the p-values, we fail to reject the null hypothesis that the lower tail dependence is statistically equal to the lower tail dependence.

We estimate the time-varying copula models for the standardized residuals. We use the GAS model for the time-varying copula distribution. The GAS model is specified in equation (10) which includes the time-varying copula parameter (δ_t) and a forcing variable from the standardized score (S_t).

$$f_t = h(\delta_t) \Leftrightarrow \delta_t = h^{-1}(f_t) \quad (10)$$

where

$$f_{t+1} = \omega + \beta f_t + \alpha I_t^{-1/2} s_t$$

$$s_t \equiv \frac{\partial}{\partial \delta} \log c(U_{1t}, U_{2t}; \delta_t)$$

$$I_t \equiv E_{t-1}[s_t s_t'] = I(\delta_t)$$

We refer to the previous estimation results for the constant copula models in choosing a time-varying copula model. The parameter estimates of the time-varying Student's t copula model are presented in Table 6.

〈Table 6〉 Estimation Results for the Time-Varying Copula Model:
Student's t GAS Model

	Parametric	
	Naïve	MSML
$\hat{\omega}$	0.0953	
	0.0114	0.3067
$\hat{\alpha}$	0.0352	
	0.0082	0.4358
$\hat{\beta}$	0.2843	
	0.0514	0.0000
$\hat{\nu}^{-1}$	0.0595	
	0.0098	0.9796
$\log L$	49.8628	

Note: The table presents the parameter estimates and the standard errors of the time-varying Student's t copula model. The standard errors of the estimated parameters are calculated from the naïve and multi-stage maximum likelihood method.

4. Risk Measures

We have shown in Figure 10 that the simulated returns calibrated to the GPD margins and a normal copula with a correlation coefficient of 0.18 replicates the actual data closely. The linear correlation coefficient between the two stock returns is computed at 0.174. Compared to the scatter plot of the actual returns in the right panel of Figure 10, however, the simulated returns are more densely populated in the middle of the distribution and do not replicate the actual returns at the extreme quantile. This is partly due to the fact that the normal copula used in the simulation process with the GPD margins does not allow either lower and upper tail dependence except for the case $\delta = 1$. As we have shown in the previous section, however, the existence of lower and upper tail dependence is evidenced by the data. So, it is

desirable to create a bivariate distribution from marginal distributions and a copula which appropriately explains the statistical characteristics of the actual data. We generate a bivariate distribution constructed from the GPD margins and a copula that can capture positive dependence of the observed data. We use both the empirical marginal models and the GPD function for estimating the tails of the marginal distributions. For the copula functions of the procedure, we employ the most heavily used seven copulas in the extant literature. We generate the bivariate joint density function via two GPD margins and a copula function, that is $f_{XY}(x, y) = f_x(x)f_y(y)c(F_x(x), F_y(y))$.

We generate 1,000,000 observations from the bivariate joint distribution and calculate the 5% and 1% VaR. We set the portfolio weights on the KOSPI 200 and the S&P 500 as 50% each. For the benchmark model with the normal marginal distributions and the Gaussian copula function, the margins are assumed as $X \sim N(0.0002897, 0.0002898)$ and $Y \sim N(0.0001061, 0.0001671)$. We presume a various levels of dependence for the Gaussian copula. In calculating the risk measures, we use numerical approximation procedures. From the simulated returns of the portfolio, we compute the daily $(1 - q) \times 100\%$ VaR and the corresponding ES.

Table 7 reports the evaluation results of the VaR and ES by simulation based on the GPD and a variety of copula distribution. According to Table 7, the daily loss on the portfolio is as low as -1.88% with 5% probability when the Clayton copula is used. If we generate the data from the Clayton copula which only allows lower tail dependence, we would observe pairs of simultaneous negative returns more with higher probability than pairs of simultaneous positive returns predicted by the bivariate normal distribution with a similar correlation coefficient. When

the daily negative return is less than -1.88% , the average loss on the portfolio is -2.83% .

〈Table 7〉 95% VaR_q and ES_q Computation Results by Simulation with Calibrated GPD Margins to Actual Data and a Parametric Copula

Copula model	VaR (%)	ES (%)
Gaussian	1.8628	2.7028
Gumbel	1.8103	2.6140
BB1	1.8712	2.7861
BB3	1.8760	2.7915
BB5	1.8201	2.6228
BB7 (Joe-Clayton)	1.8666	2.7855
Frank	1.8395	2.6307
Clayton	1.8797	2.8337
Joe	1.7709	2.5407

Note: The table reports the VaR and ES under the 95% quantile. We assume the GPD as the two margins and the nine, most heavily used copulas in the extant literature. Under the GPD margins and the Gaussian copula, for example, the 95% VaR is -1.86% and the ES is -2.70% . Under the alternative Clayton copula and the GPD margins, the 95% VaR is evaluated at -1.88% and the ES at -2.83% . The 95% VaR and ES are the highest in the case of the Clayton copula.

Table 8 reports the VaR and ES under the 99% quantile. We use the GPD as the two margins and the nine, most heavily used copulas in the extant literature. Under the GPD margins and the Gaussian copula, for example, the 99% VaR is -3.19% and the ES is -4.11% . Under the alternative Clayton copula and the GPD margins, the 99% VaR is evaluated at -3.38% and the ES at -4.49% . The 99% VaR and ES are the highest in the case of the Clayton copula. If we erroneously use the standard normal distribution for the daily losses in calculating the risk measures, we would underestimate the Value-at-Risk and expected shortfall. As can be seen from Table 9, the VaR and ES from the benchmark model underestimate the levels of risk regardless of the measures of dependence δ .

〈Table 8〉 99% VaR and ES Computation Results by Simulation with Calibrated GPD Margins to Actual Data and a Parametric Copula

Copula model	VaR (%)	ES (%)
Gaussian	3.1913	4.1055
Gumbel	3.0848	3.9605
BB1	3.3234	4.3382
BB3	3.3198	4.3515
BB5	3.1018	3.9561
BB7 (Joe-Clayton)	3.3104	4.3683
Frank	3.0958	3.9350
Clayton	3.3761	4.4922
Joe	2.9926	3.8175

Note: The table reports the VaR and ES under the 99% quantile. We assume the GPD as the two margins and the nine, most heavily used copulas in the extant literature. Under the GPD margins and the Gaussian copula, for example, the 99% VaR is -3.19% and the ES is -4.11%. Under the alternative Clayton copula and the GPD margins, the 99% VaR is evaluated at -3.38% and the ES at -4.49%. The 99% VaR and ES are the highest in the case of the Clayton copula.

〈Table 9〉 VaR and ES Computation Results by Simulation with Normal Margins and the Gaussian Copula

q	VaR (%)	ES (%)
Panel A	$\delta = 0.1$	
0.95	1.3954	1.7571
0.99	1.9816	2.2766
Panel B	$\delta = 0.5$	
0.95	1.6325	2.0530
0.99	2.3164	2.6571
Panel C	$\delta = 0.9$	
0.95	1.8373	2.3109
0.99	2.6078	2.9942

Note: The table reports the VaR and ES under the 95% and 99% quantile. We assume the normal distribution as the two margins and the Gaussian copula to generate the bivariate joint distribution. We also differentiate the correlation coefficient of the Gaussian copula, $\delta = 0.1, 0.5, 0.9$. Under the normal margins and the Gaussian copula, for example, the 99% VaR are -1.98%, -2.32%, and -2.61%, respectively. The 99% ES are -2.28%, -2.66%, and -2.99%, respectively. Under the alternative Clayton copula and the GPD margins, the 99% VaR is evaluated at -3.38% and the ES at -4.49%. Still, the 99% VaR and ES are higher in the case of the Clayton copula.

5. Out-Of-Sample Predictability Test

We evaluate the relative merit of out-of-sample forecasting ability of the alternative models based on a loss function. To that end, we introduce Giacomini and White (2006) method which innovates Diebold and Mariano (1995) and West (1996) tests in three ways. Firstly, they address the issue of misspecification by proposing the ‘forecasting method’ which includes the forecasting model, estimation procedure, and the method to choose the data. Secondly, they provide a systematic framework for nested and non-nested model forecast. This is an important thing to note because the out-of-sample forecast evaluation due to West (1996) is not applicable to nested forecasting model. Thirdly, Giacomini and White (2006) develop a procedure for evaluating the out-of-sample forecasting model which enables semiparametric or nonparametric estimation procedure. This is the most relevant to our study because the construction of out-of-sample forecasting model is semiparametric or parametric in this paper depending on the copula function.

For the possible misspecification, we use the fixed window estimation when we make out-of-sample forecast following Patton (2013). We confirm the rationale behind this choice of estimation window from Figure 9 and Table 2. The stock index return series experience structural break and can have more reliable forecast from a fixed window estimation forecast than from an expanding window estimation forecast. Although we do not use a rolling window estimation forecast, a fixed window estimation forecast is statistically similar to a rolling window estimation forecast. As discussed in Giacomini and White (2006), these two window estimation forecasts are the same in the sense that the parameters of

the forecasting model are estimated once over the in-sample data and used for all out-of-sample prediction.

First, the parameters are estimated over a subsample of a fixed window of size τ only once and used for all h -step-ahead out-of-sample stock index return predictions. The out-of-sample forecasts are calculated for the index return, and evaluated by a loss function, such as the mean absolute error (MAE), the mean-squared error (MSE), and the root-mean-squared error (RMSE). The accuracy of the h -horizon forecasts can be evaluated by the statistics computed from the prediction errors such as:

$MAE = \frac{1}{(T-h-\tau)} \sum_{t=\tau}^{T-h} |e_{\tau+h} - \tilde{e}_{\tau+h}|$, $MSE = \frac{1}{(T-h-\tau)} \sum_{t=\tau}^{T-h} (e_{\tau+h} - \tilde{e}_{\tau+h})^2$, and $RMSE = \sqrt{MSE}$. For the loss functions, the null hypothesis of equal predictive ability of the competing models for the forecasting horizon h can be established. The construction and evaluation of forecasts is iterated moving the forecasting date one period forward until the out-of-sample is exhausted. The out-of-sample predictability comparison calls for selecting the first prediction date which divides the entire data into estimation and forecasting subsamples. The selection of the first forecasting date has to be pursuant to the balance between a bias-variance trade-off. We use the out-of-sample predictability test to evaluate the performance of the models for dependence structure of the two returns. We use 3,114 observations during the period from January 04, 1999 to January 04, 2012 to estimate the forecasting models and test the out-of-sample predictability between three competing models based on 995 observations during the period from January 05, 2012 to February 26, 2016.

We compare out-of-sample predictability of the three copula-based

competing models by calculating the out-of-sample log-likelihood. The Gaussian and the Student's t copula models are the representative symmetric copulas, while the Clayton copula is chosen as the representative asymmetric copula model. These copula models are selected based on the empirical findings in section 2.2. The out-of-sample log-likelihood values reported in Table 10 are the Kullback-Leibler distance of the forecast of the competing model from the true forecasting model. Therefore, the competing model with the higher log-likelihood value can be judged to be closer to the true forecasting model. The two copula-based forecasting models are compared pairwise. According to the test statistics, the out-of-sample predictability of the Clayton and the Student's t copula models are superior to that of the Gaussian benchmark copula model. However, the superiority in the out-of-sample predictability is statistically significant only between the Student's t copula versus the Gaussian copula. Overall, the Clayton copula model is the best out-of-sample forecasting model. The out-of-sample value of the log-likelihood for each model is also presented. This result is in line with the log-likelihood value in the last row of the table. The log-likelihood value of the Clayton copula model is the largest of the three competing models.

〈Table 10〉 Out-Of-Sample Predictability Test Results

Copula	Gaussian	Clayton	Stud t
Gaussian	-	-	-
Clayton	1.4852	-	-
Stud t	1.9784	-1.0146	-
Log Likelihood	13.7053	17.7773	15.0172

Note: This table reports the t-statistics from the out-of-sample predictability test results. The two copula-based forecasting models are compared pairwise. According to the test statistics, the out-of-sample predictability of the Clayton and the Student's t

copula models are superior to that of the Gaussian benchmark copula model. However, the superiority in the out-of-sample predictability is statistically significant only between the Student's *t* copula versus the Gaussian copula. Overall, the Clayton copula model is the best out-of-sample forecasting model. The out-of-sample value of the log-likelihood for each model is also presented. This result is in line with the log-likelihood value in the last row of the table. The log-likelihood value of the Clayton copula model is the largest of the three competing models.

III. Concluding Remarks

According to the recent progress in the field of dynamic asset allocation and risk management, the out-of-sample return predictability plays an important role. Although the assumption of multivariate normality and linear dependence structure play a pivotal role in risk management, many empirical findings have confirmed the assumptions to be invalid. If these assumptions are violated, the risk measures would underestimate, especially when the portfolio return distribution is asymmetric in the extreme tails on both sides.

In this paper we use the copula distribution function methodology that can accommodate non-Gaussian asymmetric and time-varying dependence structure of the portfolio returns. We search for the copula distribution function with asymmetric tail dependence to calculate risk measures and evaluate whether those from the Gaussian copula function underestimate or overestimate true risk measures. We also investigate the impact of asymmetric tail dependence between the portfolio returns on the out-of-sample predictability of returns.

We find the simulated returns calibrated to the GPD margins and a Clayton copula replicates the actual data not only in the middle of the distribution but also at the extreme quantiles. This is also in line with empirical results that we observe pairs of simultaneous negative returns more with higher probability than pairs of simultaneous positive returns

predicted by the bivariate normal distribution with a similar correlation coefficient.

Under the GPD margins and the Clayton copula, the 99% VaR is evaluated at -3.38% and the ES at -4.49%. The 99% VaR and ES are the highest in the case of the Clayton copula. On the other hand, if we erroneously use the GPD margins and the Gaussian copula in calculating risk measures, the 99% VaR is calculated at -3.19% and the ES at -4.11%. We would underestimate the VaR and ES. We compute the VaR and ES for the 99% and 95% quantiles and confirm that the benchmark model underestimate the levels of risk regardless of the measures of dependence.

We use the out-of-sample predictability test to evaluate the performance of the asymmetric copula models. We use 3,114 observations during the period from January 04, 1999 to January 04, 2012 to estimate the forecasting models and test the out-of-sample predictability between three competing models based on 995 observations during the period from January 05, 2012 to February 26, 2016. We compare the out-of-sample predictability of the three copula-based models. The Gaussian and the Student's *t* copula models are the representative symmetric copulas, while the Clayton copula is chosen as the representative asymmetric copula model. According to the test statistics, the out-of-sample predictability of the Clayton and the Student's *t* copula models are superior to that of the Gaussian copula model. However, the superiority in the out-of-sample predictability is statistically significant only between the Student's *t* copula versus the Gaussian copula. Overall, the Clayton copula model is the best out-of-sample forecasting model.

References

- Bouyé, E., N. Gaussel, and M. Salmon, "Investigating Dynamic Dependence Using Copulae," *Working Papers WP01-03*, Financial Econometrics Research Centre, 2002.
- Bouyé, E., and M. Salmon, "Dynamic Copula Quantile Regressions and Tail Area Dynamic Dependence in Forex Markets," *European Journal of Finance*, Vol. 15, 2009, pp. 721-750.
(DOI: <http://dx.doi.org/10.1080/13518470902853491>)
- Chen, X., Y. Fan, and A. Patton, "Simple Tests for Models of Dependence between Multiple Financial Time Series with Applications to U.S. Equity Returns and Exchange Rates," *Working Paper WP04-19*, Financial Econometrics Research Centre, 2004.
(DOI: <http://dx.doi.org/10.1016/j.jeconom.2005.03.004>)
- Chen, X., and Y. Fan, "Estimation of Copula-Based Semiparametric Time Series Models," *Journal of Econometrics*, Vol. 130, 2006, pp. 307-335.
(DOI: <http://dx.doi.org/10.1016/j.jeconom.2005.03.004>)
- Diebold, F. X., and J. A. Lopez, "Forecast Evaluation and Combination," in *Handbook of Statistics*, Vol. 14: Statistical Methods in Finance, ed. by G. S. Maddala, and C. R. Rao. Amsterdam: North-Holland, 1996, pp. 241-268.
- Frahm, G., M. Junker, and R. Schmidt, "Estimating the Tail-Dependence Coefficient: Properties and Pitfalls, Insurance," *Mathematics and Economics*, Vol. 37, 2005, pp. 80-100.
(DOI: <http://dx.doi.org/10.1016/j.insmatheco.2005.05.008>)

- Genest, C., and R. J. MacKay, "Copulas Archimediennes et Familles de lois Bidimensionnelles dont les Marges sont Donnees," *Canadian Journal of Statistics*, Vol. 14, 1986, pp. 145-159.
(DOI: <http://doi.org.ssl.openlink/10.2307/3314660>)
- Genest, C., K. Ghoudi, and L.-P. Rivest, "A Semiparametric Estimation Procedure of Dependence Parameters in Multivariate Families of Distributions," *Biometrika*, Vol. 82, No. 3, 1995, pp. 543-552.
(DOI: <http://dx.doi.org/10.1093/biomet/82.3.543>)
- Giacomini, R., and H. White, "Tests of Conditional Predictive Ability," *Econometrica*, Vol. 74, 2006, pp. 1545-1578.
(URL: <http://www.jsor.org/stable/4123083>)
- Nelson, R. B., *An Introduction to Copulas*, Lecture Notes in Statistics, Springer-Verlag: New York, 1999.
- Nelson, R. B., *An Introduction to Copulas*, 2nd Ed., Springer, New York, 2006.
- Patton, A. J., "Estimation of Copula Models for Time Series of Possibly Different Lengths," *Working paper 2001-17*, University of California, San Diego, 2001a.
- Patton, A. J., "Modelling Time-Varying Exchange Rate Dependence Using the Conditional Copula," *Working paper 2001-09*, University of California, San Diego, 2001b.
- Patton, A. J., *Applications of Copula Theory in Financial Econometrics*, Unpublished Ph.D. dissertation, University of California, San Diego, 2002.
(URL: <http://www.riss.kr/pdu/ddodLink.do?id=T10574896>)
- Patton, A. J., "On the Out-Of-Sample Importance of Skewness and Asymmetric Dependence for Asset Allocation," *Journal of*

Financial Econometrics, Vol. 2, 2004, pp. 130-168.

Patton, A. J., "Estimation of Multivariate Models for Time Series of Possibly Different Lengths," *Journal of Applied Econometrics*, Vol. 21, 2006a, pp. 147-173.

(DOI: <http://dx.doi.org/10.1002/jae.865>)

Patton, A. J., "Modelling Asymmetric Exchange Rate Dependence," *International Economic Review*, Vol. 47, 2006b, pp. 527-556.

(URL: <http://www.jstor.org/stable/3663514>)

Patton, A. J., "Copula-Based Models for Financial Time Series," in T.G. Andersen, R.A. Davis, J.-P. Kreiss and T. Mikosch, *Handbook of Financial Time Series*, Springer Verlag, 2009a.

Patton, A. J., "Are "Market Neutral" Hedge Funds Really Market Neutral?," *Review of Financial Studies*, Vol. 22, No. 7, 2009b, pp. 2495-2530.

Patton, A. J., "A Review of Copula Models for Economic Time Series," *Journal of Multivariate Analysis*, Vol. 110, 2012, pp. 4-18.

(DOI: <http://dx.doi.org/10.1016/j.jmva.2012.02.021>)

Patton, A. J., "Copula Methods for Forecasting Multivariate Time Series," *Handbook of Economic Forecasting*, Vol. 2, 2013, pp. 899-960.

West, K. D., "Asymptotic Inference about Predictive Ability," *Econometrica*, Vol. 64, 1996, pp. 1067-1084.

(URL: <http://www.jstor.org/stable/2171956>)

Zivot, E., and J. Wang, *Modeling Financial Time Series with S-PLUS*, Springer, 2006.

포트폴리오 수익률 분포의 비대칭적 의존성의 표본외 예측가능성: Copula 분포함수에 의한 추정*

이호진**

〈요 약〉

주식수익률의 표본외 예측가능성은 동태적 자산배분과 포트폴리오 위험관리에 중요한 역할을 한다. 위험관리에서 다변량 정규분포의 가정과 주식수익률간 선형의존성 가정이 주로 이용되는 반면, 실제 데이터에서 다변량 정규분포와 선형의존성 가정은 위배되는 것이 일반적이다. 본 연구는 포트폴리오를 구성하는 주식수익률간 꼬리부분의 비대칭 의존성과 비정규성을 코플라 분포함수로 모형화하고 포트폴리오를 구성하는 자산의 한계분포는 일반화 파레토분포로 모형화하여 KOSPI 200과 S&P 500으로 구성된 지수포트폴리오의 위험척도를 측정하였다. 일반화 파레토분포와 가우스 코플라함수를 결합하여 측정하는 모형은 일반화 파레토분포와 Clayton 코플라함수를 이용하는 모형에 비해 지수 포트폴리오의 위험을 과소추정하는 것으로 나타났다. 또한 지수포트폴리오의 꼬리부분의 비대칭 의존성이 표본외 예측력에 미치는 영향을 평가하였는데, Clayton 코플라함수에 의한 지수포트폴리오 수익률의 표본외 예측력이 가장 뛰어난 것으로 나타났다. Clayton 코플라함수는 다변량 수익률분포에서 왼쪽 꼬리분포간의 비대칭 의존성을 반영하는, 즉 포트폴리오 구성자산들에서 동시에 양의 수익률이 나타나는 경우에 비해 동시에 음의 수익률이 나타나는 경우가 확률적으로 높은 사실을 반영하는, 통계적 특성을 가진 코플라함수이다. 본 연구는 위험척도 추정과 수익률의 표본외 예측력 검증에서 Clayton 코플라함수가 우월한 것은 이러한 통계적 특성에 기인한 것임을 실증하였다.

핵심주제어: 코플라함수, 비대칭 의존성, 꼬리분포 의존성, 표본외 예측, VaR, 기대손실
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